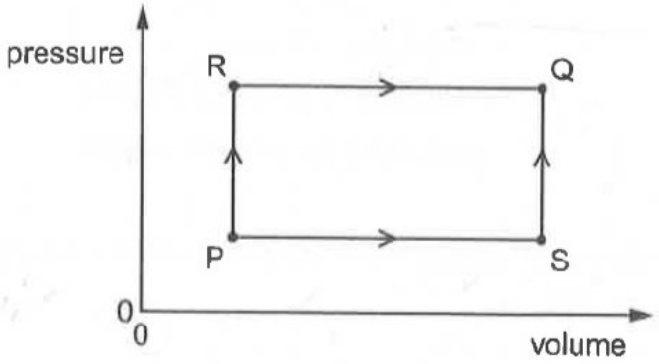
17 A fixed mass of gas undergoes changes of pressure and volume as shown.



When a gas is taken from state P to state Q by the stages PR and RQ, there is a thermal transfer of 8 J of heat into it and 3 J of work are done by it. When the resultant change is achieved by stages PS and SQ, 1 J of work is done by the gas.

In this case, what is the thermal transfer of energy?

A 6 J out of the gas

B 6 J into the gas

C 10 J into the gas

D 12 J out of the gas